# Lewis Structures

Valence electrons, multiple bonds, formal charge and resonance

Name _____ Class _____ Date _____

**USEFUL INFORMATION**

Valence electrons: group number for groups 1–2; group number minus 10 for groups 13–18

Add 1 electron per negative charge; subtract 1 per positive charge

Formal charge = valence $e^-$ − lone-pair $e^-$ − ½ bonding $e^-$

Draw every lone pair. Hydrogen is always an outer atom.

## A. Counting electrons

**1.** Count the total valence electrons in (a) $CH_4$ (b) $PCl_3$ (c) $NH_4^+$ (d) $CO_3^{2-}$

## B. Drawing structures

**2.** Draw Lewis structures for (a) $H_2S$ (b) $NF_3$ (c) $CCl_4$

**3.** Draw Lewis structures for (a) formaldehyde, $CH_2O$ (carbon is central) (b) ethene, $C_2H_4$ (the two carbons are bonded to each other)

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**4.** Draw the Lewis structure of carbon disulfide, $CS_2$. How many lone pairs are there in the whole molecule?

**5.** Draw the Lewis structure of the ammonium ion, $NH_4^+$, and calculate the formal charge on nitrogen.

## C. Formal charge and resonance

**6.** Ozone, $O_3$, has 18 valence electrons. Draw one Lewis structure and calculate the formal charge on each oxygen atom.

**7.** Draw the resonance structures of the nitrate ion, $NO_3^-$. What does resonance tell you about the three N–O bond lengths?

## D. Exceptions

**8.** Explain why $BF_3$ is an exception to the octet rule. How many electrons surround boron?

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## E. Challenge

**9.** Xenon tetrafluoride, $XeF_4$, has 36 valence electrons. Draw its Lewis structure. How many electrons surround xenon, and why is this possible?

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